



STONE CLADDING SYSTEM
(MECHANICAL FIXING SYSTEM)



INTRODUCTION

General:

Facades can be fixed in different ways. The most reliable and effective way is the mechanical fixing. In this way, we use the steel systems to support and fix the required facing elements such as stone, marble, granite etc.

MECHANICAL FIXING COMARED TO CONVENTIONAL FIXING :

Mechanical Fixing	Conventional Fixing
1. Time Saving	1. Time Consuming
2. Labor Saving	2. Labor Saving
3. Design: Staggered or stacked	3. Desing: Staggered
4. Straightness: Easy to adjust when mistakes arise due to adjustability features	4. Straightness: Difficult to correct
5. Maintenance: Easy maintenance	5. Maintenance: Require extensive work

Analysis of Loads:

Dead Load: Weigh of Stone / Marble

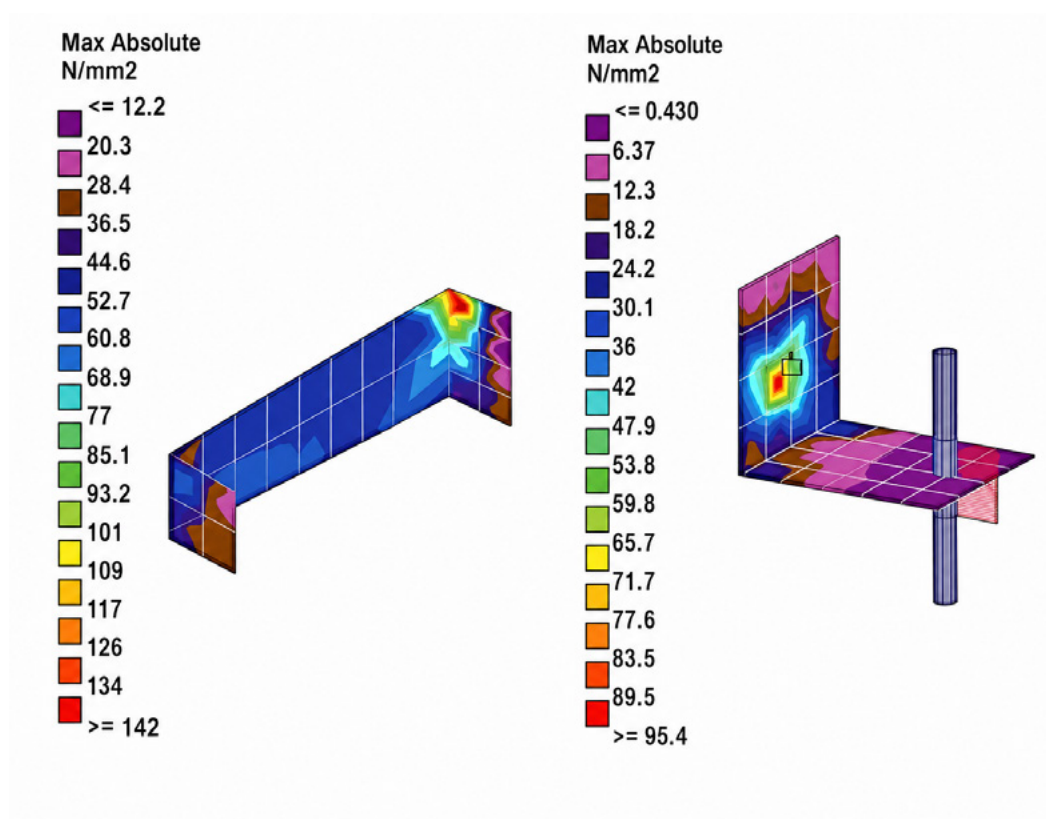
Wind Load: For wind speed as per QCS 2014, In accordance to ASCE7-10 or BS 6399.

Seismic Load: In accordance to ASCE7-10 or UBC 1997.

Thermal Loads: For Thermal Expansion of Stone.

Design Steps:

- 1- Designing Section by considering combination of loads for Limit State of Flexure, Shear & Tension.
- 2- Comparing the stresses with allowable strength of sections based on the yield stress of materials.
- 3- Checking for Limit state of Serviceability, Governed by Deflection criteria.



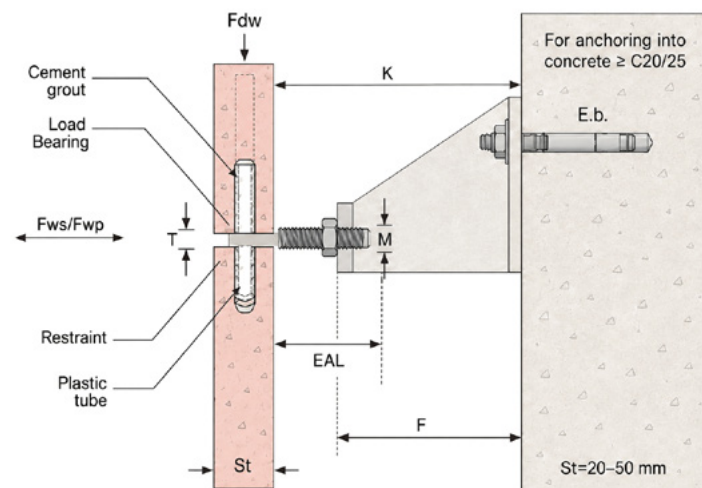
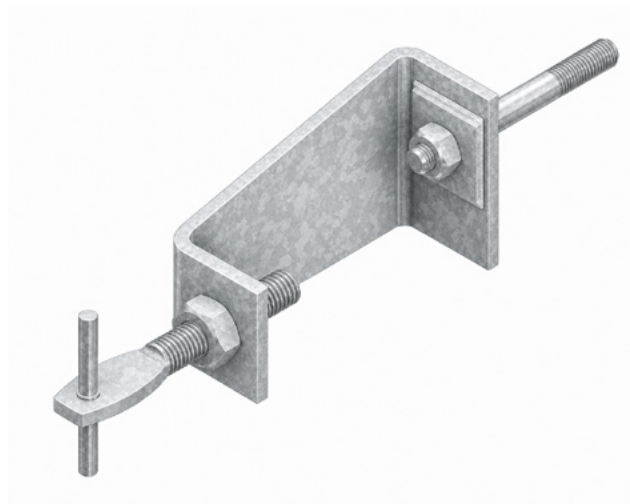
Material Specification:

Material complying with ASTM A240/ ASTM A666/BS EN 10088-304.316.316L standards.

Direct Fixing: CRV Bracket (C-Bracket)

C- Bracket:

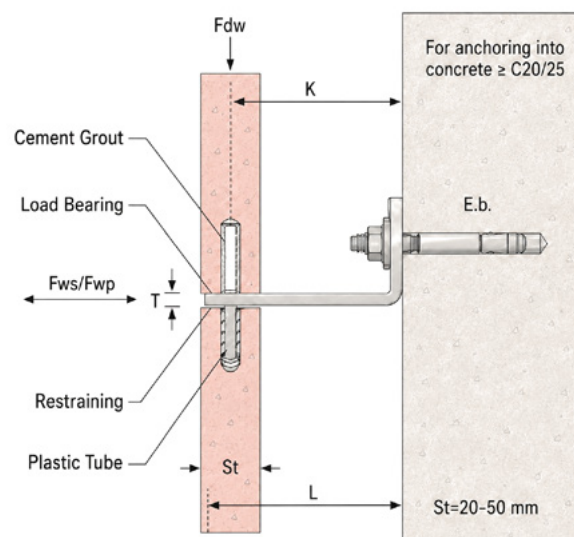
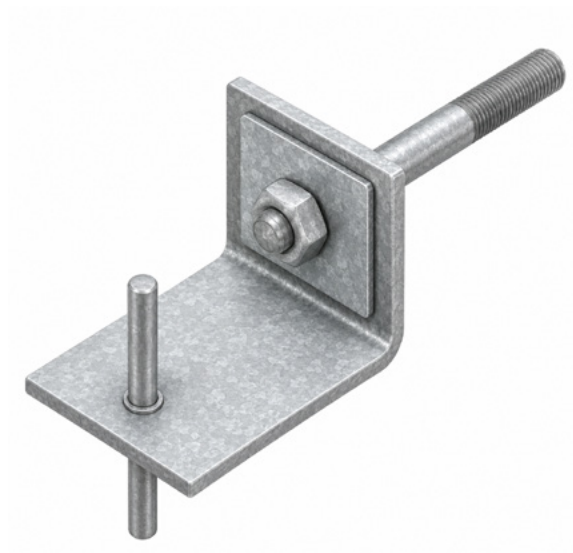
- Heavy duty bracket used in bigger cavity distance from the wall or heavy cladding.
- Loads are submitted based on full design.
- Adjustability up to ± 25 mm



Direct Fixing Angle Bracket (L-Bracket)

L- Bracket:

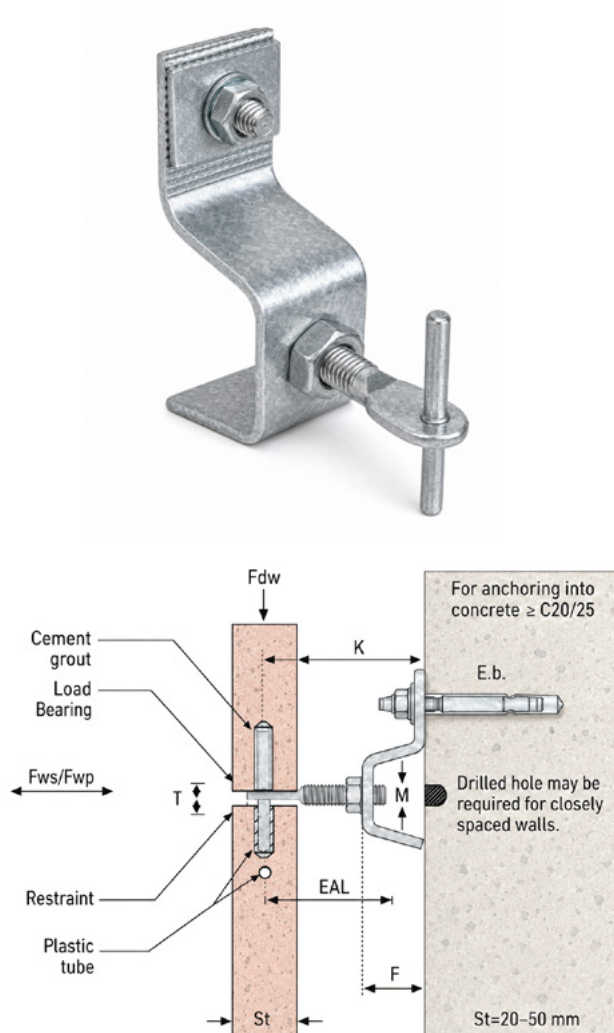
- Used generally for small cavity
- Loads are submitted based on full design.
- Limited adjustability
- Easy installation, economical



Direct Fixing Z-Bracket with returned leg

Z- Bracket:

- Durable material construction for long-lasting performance.
- The support and restraint brackets are adjustable in 2 directions.
- Loads are submitted based on full design.
- Compatibility with various adjustability & cavities 600 – 1200mm.
- Designed for easy installation and secure stone cladding support.
- Suitable for both interior and exterior applications.





DELIVERY, HANDLING AND STORAGE:

- Delivery: Materials are to be delivered in original packages, containers or bundles containing the manufacturers or supplier's identification.
- Transportation: In transportation or lifting, handle products with care avoiding excessive stress and preventing damages.
- Storage: Store materials under a shaded area and keep them dry and protected against damage due to weather condition, humidity level, job site location, storage condition and surface contamination.

Quality Assurance:

Our Quality Assurance Program has been developed and implemented for compliance with ISO9001: 2008. We also comply with various industry standards and specifications.

Round Washer DIN 440, DIN 9021

Washers (RW) | DIN 440 | DIN 9021:

Washer Dimension	Outer Diam. (mm)
M6	22
M8	24
M10	30
M12	45
M12	37
M16	50



Hexagon Nut (HN) | DIN 934 or DIN EN 24032 :

Nut Thread Dimension	S/m ISO (mm)
M6	10/6
M8	13/7.5
M10	16/9.5
M12	18/12



Hexagon Bolt (HB) | DIN 933 or EN 24017:

Diameter (mm)	Length (mm)
M6	50/60/70
M8	50/60/80/100
M10	80/100/120
M12	80/100/120



• Finishing available from washer, nut & bolt: Galvanized, SS 304 & SS 316.

Flanged Pin/Pin

General:

Special Flanged pin is used for the top & bottom bracket where an approximate between 5 and 6mm inserted into the stone panel for support and resistance effects.

Specification	Sizes (mm)
Length (mm)	50 - 60
Pin Diameter (mm)	4, 5, 6

- Finishing available for Pin: Galvanized, SS 304 & SS 316.



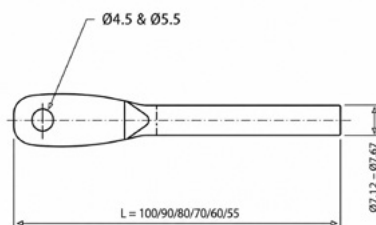
Extension Arm

General:

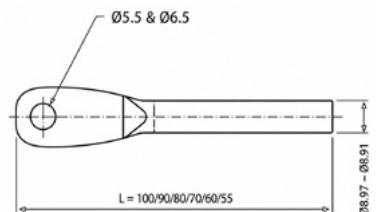
It is a standard arm which comes in various diameter & length as per suitability. Extension arm allows for cavity adjustment also allows an easy mounting in conjunction with the spring nut & pin.

Types:

1. EA M8



2. EA M10



- Finishing available for Extension Arm: Galvanized, SS 304 & SS 316.

Product Data Sheet

K5-Mechanical Fixation Access

REV 00, July 2026

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